AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 33 · STUDENT LESSON

Model Cards and Human Review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Trace the claim. Mark the unknown. Record the review.

Write model cards whose versions,metrics and use limits match their actual evidence.

Keep one habit: Complete paper card → pictured source-count limits → twelve-pair audit → two-card clinic → independent unsupported claim.

MAIN LESSON · SECTION 1

Can a reader trace this model claim to its evidence?

> A model card records supported use,limits and review status;it is not a certificate.

Learning goals

I can write a **model card** whose claims match **evidence**.

I can separate **intended use** from unsupported use.

I can review a card for missing evaluation details.

Recall what an evaluation must preserve

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Over**fitting**. Give an example and a mistake that the idea helps you avoid.
- What should remain visible after an evaluation?

End goal

By the end, I can explain **model card**s and **human review** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A **model card** defines **intended use**, **evidence**, limitations, risks, decision boundaries, monitoring, and human escalation.



LEARNING TARGET Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Link each statement to a retained record or an explicit unknown

> Do not fill missing fields with optimistic language.

Model-card purpose and fictional scope

A **model card** helps another person understand what was built and what **evidence** exists. It is a record, not a certificate of safety. Link statements to saved data and run records.

Use unknown or not evaluated where **evidence** is absent instead of filling gaps with optimistic language.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **model card**s and review.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use **model card**s and **human review** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Read a complete bounded card before reviewing its headline

> Unknowns can be documented honestly without making the model ready for a real use.

Complete paper card and its supplied metrics

Complete main paper model-card field	Evidence-linked entry / status
Identity / use	Classroom parcel-lane classifierV1; fictional clear/blocked method-comparison record only.
Data / split premise	Training40, validation 20, test 12 with separate source groups as supplied; detailed training files and actual code state are not supplied.
Positive / metrics	Blocked positive, clear/blocked axes; TN 7, FP 1, FN 2, TP 2; accuracy 9/12=75%, precision 2/3≈66.7%, recall 2/4=50%.
Baseline	Previously fitted always-clear baseline 8/12≈66.7%on the same references; training label counts are not supplied.
Unresolved fields	Exact feature/preprocessing/code/dependency versions, training provenance, collection coverage and teacher review status are not supplied. Mark unknown/request evidence.
Excluded / unsupported use	No real parcel, student, agricultural or professional diagnosis decision is supported.
Review / update	Draft teaching card; no real sign-off is asserted. Version changes require matching evidence and renewed review record.

Example card: classroom parcel classifier v 1; purpose: compare toy classification methods on fictional clear/blocked labels. **Training** 40, validation 20, test 12 with separate record groups.

Frozen test TN 7 FP 1 FN 2 TP 2, blocked positive: accuracy 9/12, precision 2/3, recall 2/4. Always-clear baseline scores 8/12. No real parcel or student decision is supported.

Trace the evidence behind the statements

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

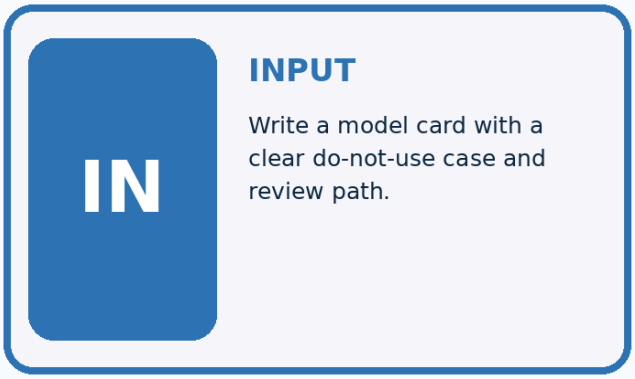
Keep the distinction between an observed result and a proposed next step.

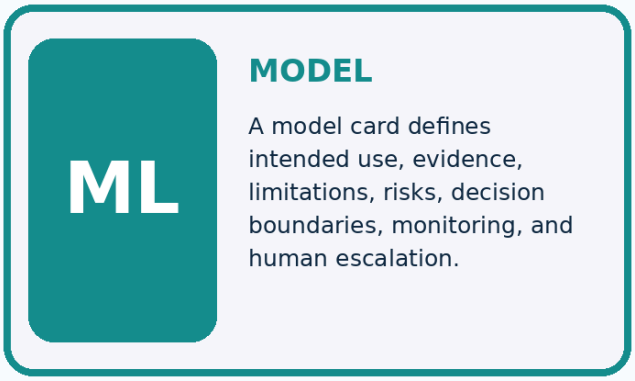
▼ Earlier wording, alternatives and picture transcripts

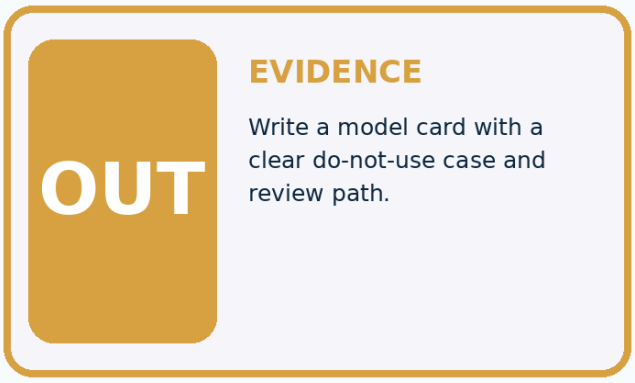
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Write a **model card** with a clear **do-not-use** case and review path.

MODEL IDEA A **model card** defines **intended use**, **evidence**, limitations, risks, decision boundaries, monitoring, and human escalation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD **Intended use**: classroom exploration of visible leaf spotting. **Evidence**: test metrics, confusion matrix, subgroup slices, and data version.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

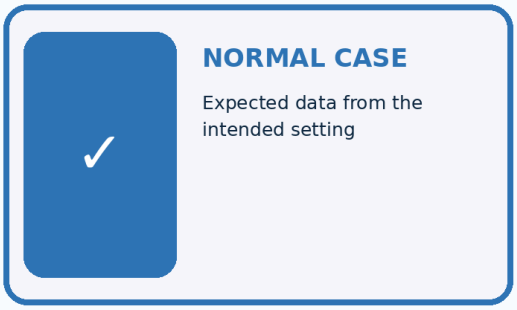
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

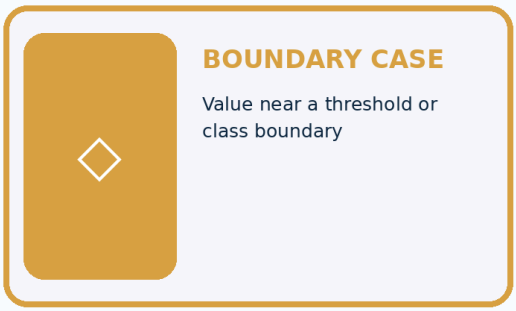
Compare an attractive claim with a traceable model record

> Good wording does not create missing evaluation evidence.

Four evaluation conditions



NORMAL Expected data and expected behavior.

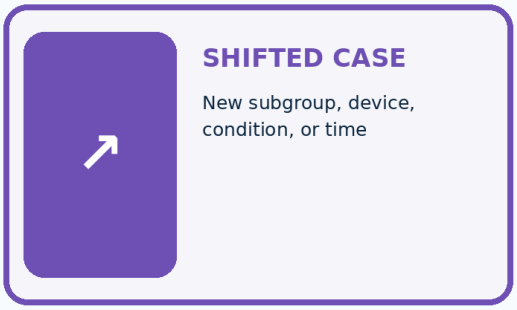


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The publicity sentence becomes “**80%** accurate for every plant”.

▼ **Check after predicting**

Prediction: Reject that unbounded statement.

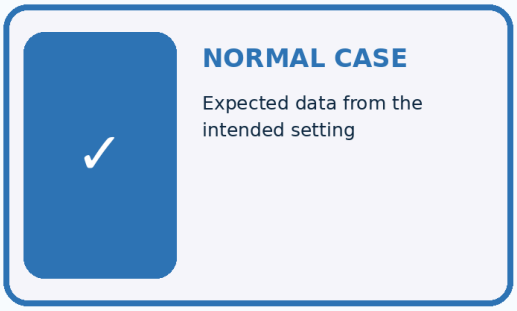
Reason: The measured rate belongs to twenty related indoor photos, not every plant or setting.

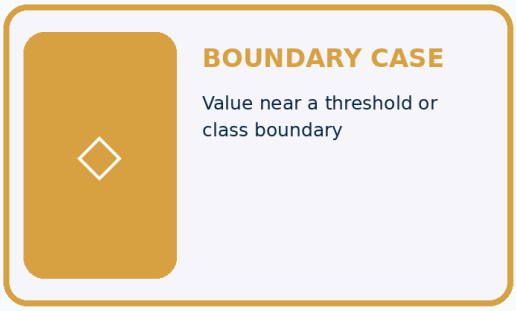
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

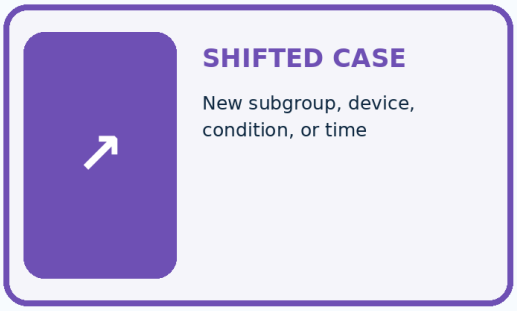
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Keep source identities and collection conditions beside the headline

> Twenty related views are not twenty independent plants.

Complete pictured facts and separate-card scope

Write a **model card** whose performance statement can be traced to a concrete evaluation.

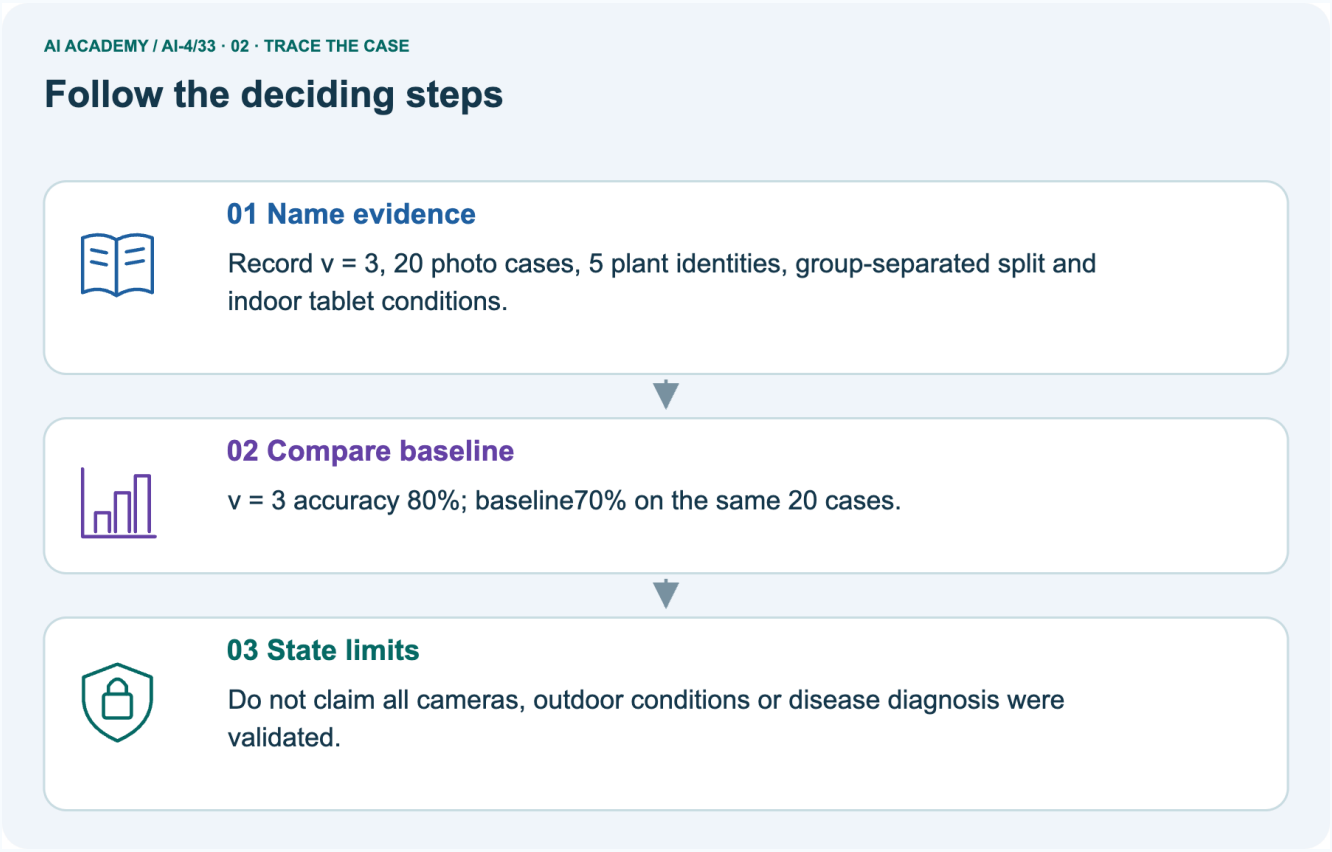
Toy v = 3 classifier scored16/20 on five unseen plants with four views each, under indoor tablet lighting. Baseline scored14/20.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main cardV1 has12 test records and matrixTN7/FP 1/FN 2/TP 2; baseline 8/12. Separate pictureV3 has16/20 photo cases correct, baseline 14/20, and only 5 unseen plant identities with 4 views each under indoor tablet conditions. Its80%versus 70%accuracy does not provide a confusion matrix, positive recall or performance on every plant.

A bounded sentence must retain version, 20 photo/5 plant counts, conditions and the unknown broader coverage. No original model/code artifacts or actual teacher approval are supplied by these worked numbers.

Trace version,evidence and bounded use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name evidence	Record v = 3, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.
2. Compare baseline	v = 3 accuracy 80%; baseline70% on the same 20 cases.
3. State limits	Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

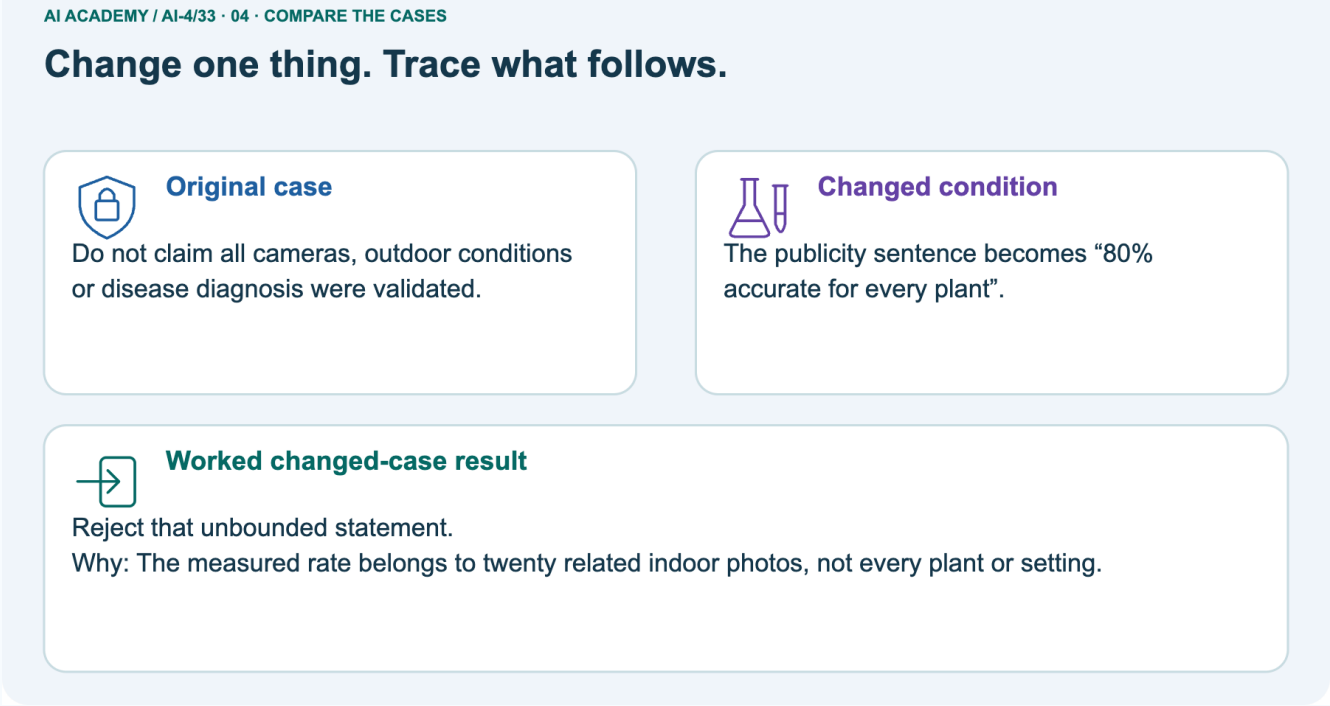
Read the picture in order. Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the “every plant”publicity revision

- Change one thing: The publicity sentence becomes “**80%** accurate for every plant”. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain review and certification limits

- A **model card** communicates **evidence**; it is not certification of safety, fairness or suitability for consequential plant decisions.

- Read the labelled picture: Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main metric verification

Model cards and review at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Purpose and version	Toy classifier v 1	Fictional data only
Evidence	12 held-out cases	Full matrix and baseline
Limits and review	No real decisions	List unresolved coverage

The model card should name feature availability, preprocessing fitted on training, model settings, reference-label rules and test date. Include subgroup coverage if evaluated; otherwise say it was not assessed. Describe relevant failure examples without exposing personal information.

Verify the worked accuracy, precision and recall. Explain the steps or evidence behind your response.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 33 • WORKED EVIDENCE

1. Intended use: classroom exploration of visible leaf spotting.
2. Evidence: test metrics, confusion matrix, subgroup slices, and data version.
3. Do not use for pesticide, medical, or agricultural diagnosis.
4. Uncertain, unfamiliar, or high-consequence cases route to a teacher or qualified adult.

STEP 1 **Intended use**: classroom exploration of visible leaf spotting.

STEP 2 **Evidence**: test metrics, confusion matrix, subgroup slices, and data version.

STEP 3 Do not use for pesticide, medical, or agricultural diagnosis.

STEP 4 Uncertain, unfamiliar, or high-consequence cases route to a teacher or qualified adult.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

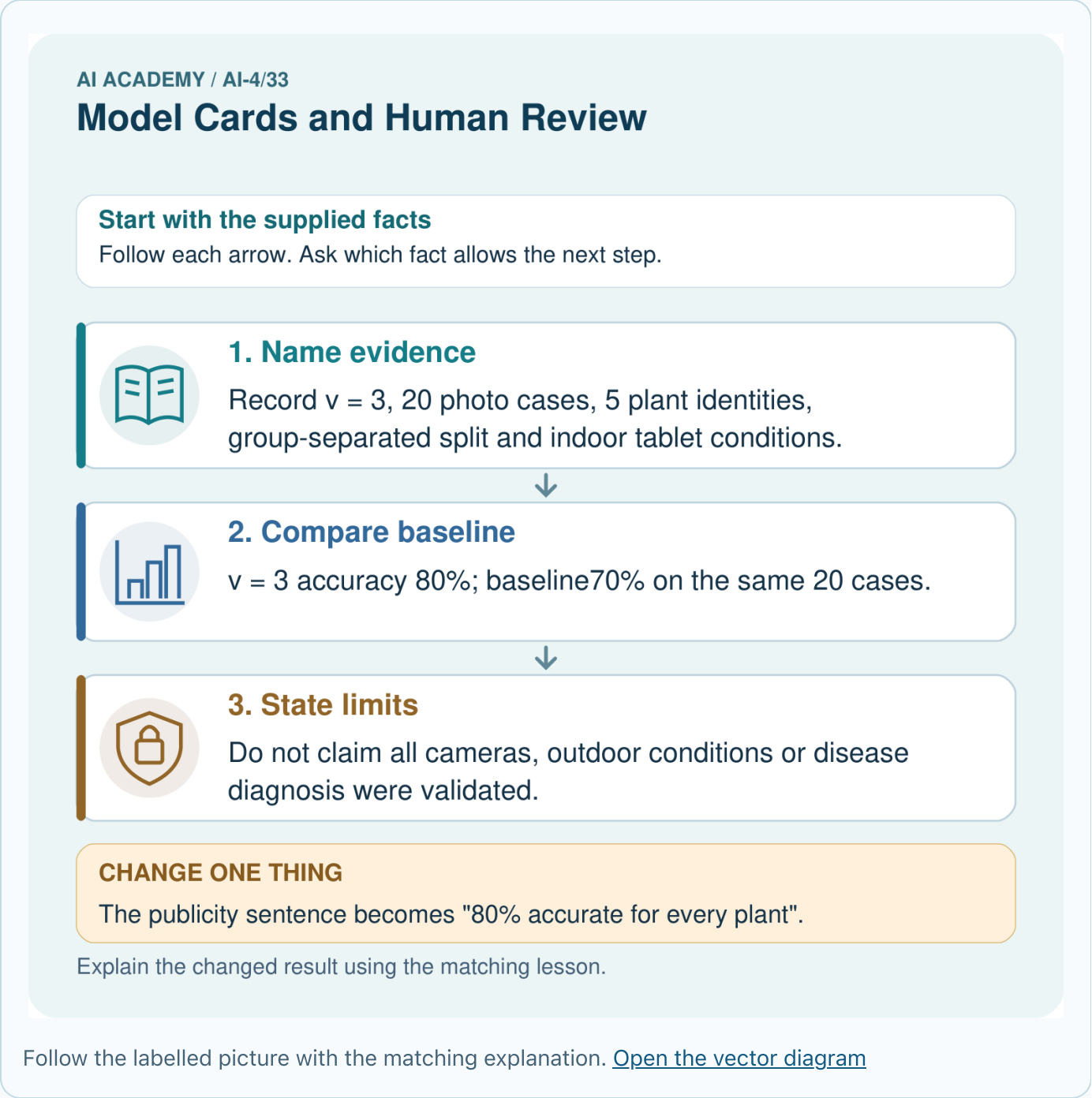
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Model Cards and Human Review

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a card field mistaken for verified evidence

> A statement needs a source and a review status,not just a heading.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE A disclaimer at the bottom of the screen is a complete safety plan.

Explain limitations and next checks

- Do not claim a synthetic classroom dataset represents every school or age group. Missing calibration, subgroup or shift **evidence** must remain explicit.

  A useful card may conclude that the model should stay a classroom demonstration.

> **What should change in a card after a threshold revision?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A disclaimer at the bottom of the screen is a complete safety plan. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise an evidence-linked peer review with less help

> Record which questions were resolved and which remain open.

Try supported arithmetic W2

Use workbook W2 for the connected example “**Model card**s and review”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Trace claims to predictions and keep updates versioned

A peer reviewer traces one headline number back to the saved **prediction**s, checks denominators and looks for mismatches between **intended use** and **evidence**. Record requested changes and whether each was resolved.

A card must be updated when its model, data or **intended use** changes.

Explain the evidence beyond a final number

- Use the worked case for **Model card**s and review. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Write the pictured bounded replacement

- Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the fresh independent90%school claim

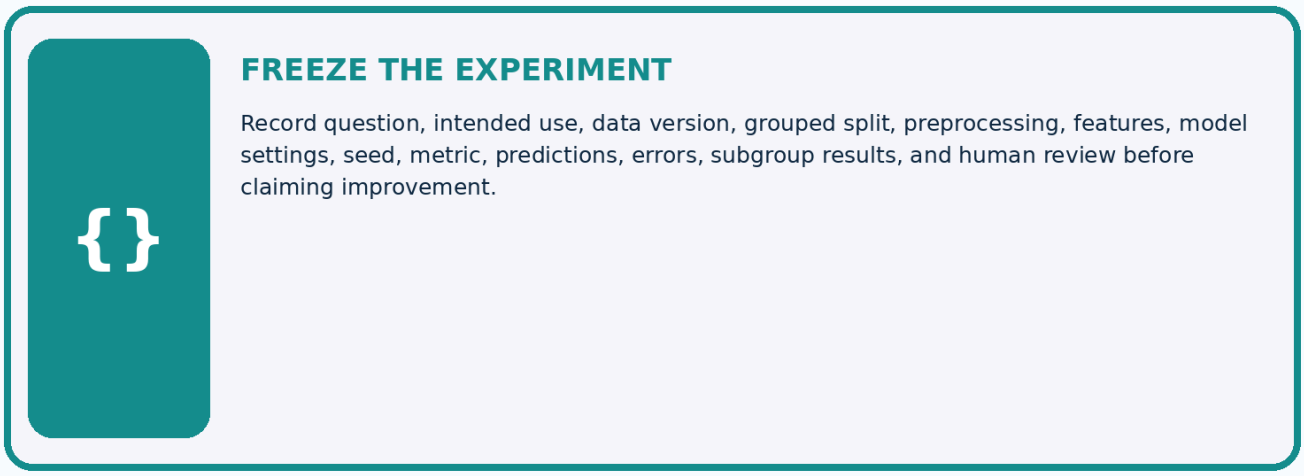
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + **intended use** + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Check arithmetic

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete two-card evidence and review clinic

> Every draft,metric source and review request needed for the promised comparison is supplied.

Complete twelve-case saved predictions

Saved test ID	Reviewed reference	FrozenV1 prediction	Fixed clear baseline
R01	clear	clear	clear
R02	clear	clear	clear
R03	clear	clear	clear
R04	clear	clear	clear
R05	clear	clear	clear
R06	clear	clear	clear
R07	clear	clear	clear
R08	clear	blocked	clear
R09	blocked	clear	clear
R10	blocked	clear	clear

Saved test ID	Reviewed reference	FrozenV1 prediction	Fixed clear baseline
R11	blocked	blocked	clear
R12	blocked	blocked	clear

Two complete cards to inspect

Complete card draft to compare · Text to inspect

> **Complete card draft to compare:** DraftA
>
> **Text to inspect:** “Parcel-laneV1is75%accurate everywhere. It beats the baseline, so it is safe for all school decisions and needs no review.” The record supplies no code/preprocessing versions or coverage beyond its fictional test.

> **Complete card draft to compare:** DraftB
>
> **Text to inspect:** “In this supplied fictionalV1 record, TN 7/FP 1/FN 2/TP 2on12 source-separated test cases give 75%accuracy, 2/3 precision and 1/2 recall, with always-clear baseline 8/12. Intended use is classroom analysis of toy clear/blocked labels. Exact code/preprocessing/training provenance and collection coverage are not supplied; review is pending. No real student/parcel decision is supported.”

Trace,rewrite and record review status

1. Model Card Clinic: compare both complete drafts, trace every quantitative statement to the twelve saved pairs, and mark each overclaim or unknown. Write a revised evidence-linked card with explicit status for each review request; do not fabricate a completed teacher approval or absent model artifacts. Distinguish a paper-teaching record from an actually reproducible classifier release.

   The project card should cite the learner’s own saved evidence, not copy this fictional model as their result. Keep the selected new 90%universal-school claim for an uncoached first correction request.

Complete review requests and readiness limits

Review request / gate · Complete evidence action and honest status

> **Review request / gate:** R1 Trace arithmetic
>
> **Complete evidence action and honest status:** UseR01–R12to recover all four cells, candidate 9/12 andbaseline8/12; record the check as done only after actually computing it.

> **Review request / gate:** R2 Bound universal use
>
> **Complete evidence action and honest status:** Replace “everywhere/all school decisions/safe”with the supplied fictional-case scope. DraftBdoes this wording repair; it does not add new model evidence.

> **Review request / gate:** R3 Check reproducible system identity
>
> **Complete evidence action and honest status:** Request actual feature/preprocessing/code/dependency/data versions and source membership from the project owner; unknown in this worked fixture.

> **Review request / gate:** R4 Review human role and unsupported inputs

> **Complete evidence action and honest status:** Specify who reviews the project, what evidence they inspect and what fallback the classroom demo uses; do not invent an already completed sign-off.

> **Review request / gate:** R5 Decide supported next action
>
> **Complete evidence action and honest status:** A complete draft can support a teacher's paper exercise review; an executable or real-use classifier release is not established by these documents. Preserve unresolved status and requests.

Normal,missing-matrix,missing-artifact and changed-version probes

Complete model-card probe	Expected bounded check
Normal mainV1 record	Candidate9/12, baseline 8/12, precision 2/3, recall 2/4; all denominators and error types retained.
Boundary one headline accuracy but no matrix	Cannot infer recall, class mix or error consequences from that number alone.
Missing code, data or collection facts	Mark not supplied and request them; do not invent model/version/coverage evidence.
Changed model, data or intended use	Version the card and link new evaluation/review status; do not inherit an old claim automatically.

Retain the evidence ledger

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Model Card Clinic: compare two cards, find unsupported claims and missing gates, then author a release-ready version.

1. Write the hypothesis and expected **evidence** before running.

2. **Model Card** Clinic: compare two cards, find unsupported claims and missing gates, then author a release-ready version.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: request evidence for a fresh universal model claim

> Do not substitute the workedV1or picturedV3numbers for missingV2evidence.

Complete independent claim and scope

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

The selectedV2 card supplies only “90%accurate”and “works for every school.” It gives no numerator, denominator, source context, split history, baseline or predictions. You cannot audit the 90%or replace it with a new invented score. Request the missing evidence and write a bounded unknown-scope statement while preserving the unverified reported claim as unverified.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 33 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A **model card** says 90% accuracy but gives no **intended use**, data conditions or human-review rule. What should be added before classroom use?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Complete the Plant Classifier **model card** and require teacher sign-off before any classroom demonstration.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project card and human review grounded in actual records

> A reader should distinguish a supported claim,a pending request and an unperformed check.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Complete the Plant Classifier **model card** and require teacher sign-off before any classroom demonstration.

Complete the project model card from actual saved data/run records:intended users/use, model and data versions, label definitions, source/split counts, metrics/matrices/baselines/slices, limitations, excluded uses, unsupported-input handling and human review status. Keep teacher sign-off as an actual recorded project action, not a claim copied from a template.

Classroom labels and performance exercises do not establish professional diagnosis or real-world suitability.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

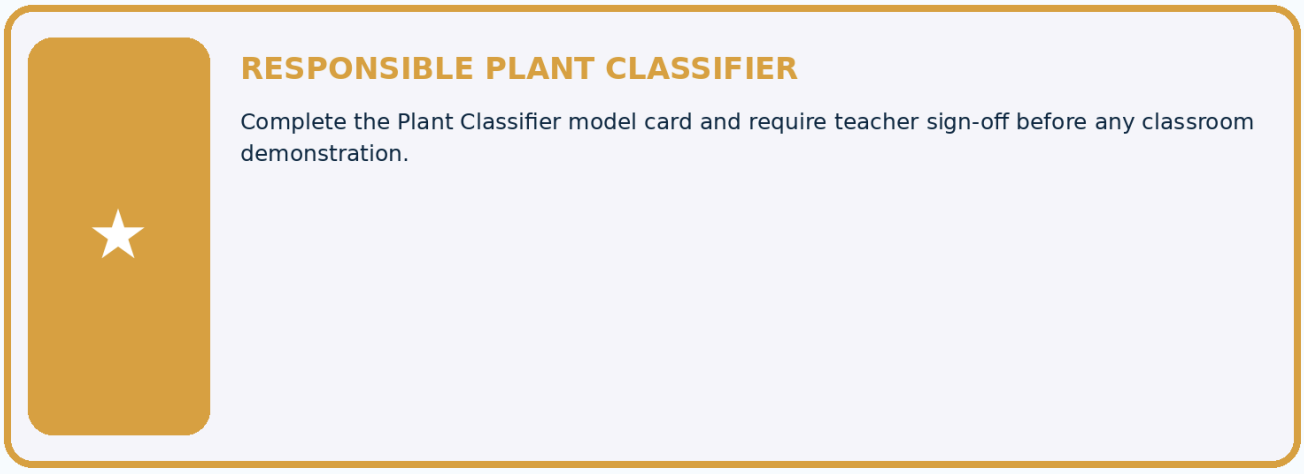
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with one traceable claim and one unresolved question

> Explain what the card communicates and what only further evidence/review can establish.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Write a **model card** whose claims match **evidence**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
model card	deployment-facing model documentation	
intended use	approved purpose and setting	
do-not-use	explicit prohibited application	
human review	authorized person checks or decides	

Closer-explanation terms

Word	Meaning
Model card	A concise structured account of a model and its limits
Intended use	The purpose and conditions the evidence supports
Out-of-scope use	A purpose for which the model is not supported

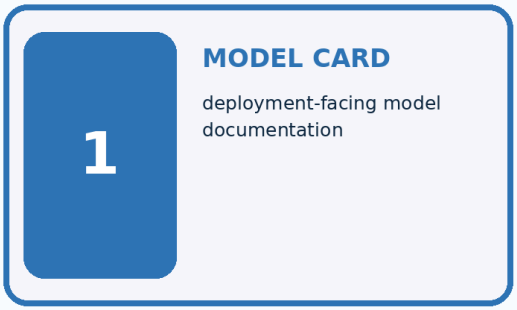
Word	Meaning
Review status	The current state of checks and unresolved issues

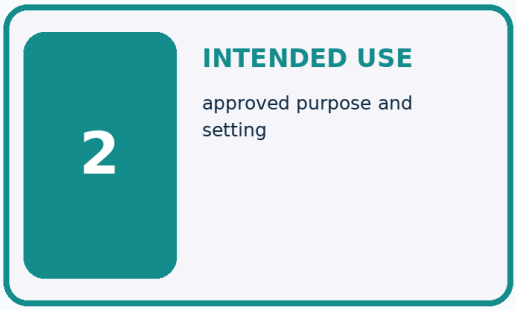
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A disclaimer at the bottom of the screen is a complete safety plan.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **model card**, **intended use**, **do-not-use**, **human review**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A disclaimer at the bottom of the screen is a complete safety plan.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State **intended use**, metric, limitation, and human responsibility.

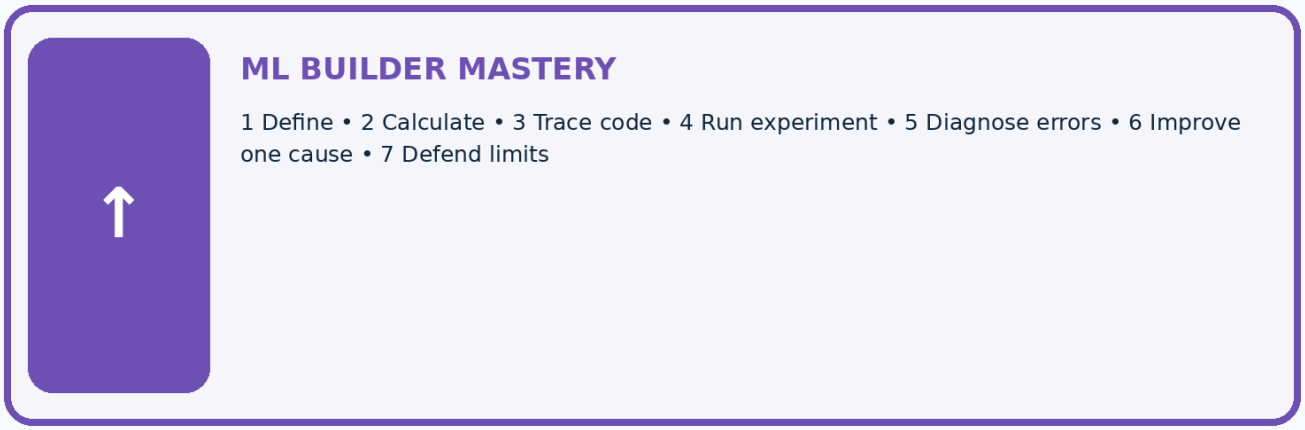
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh unsupportedV2universal-school claim for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

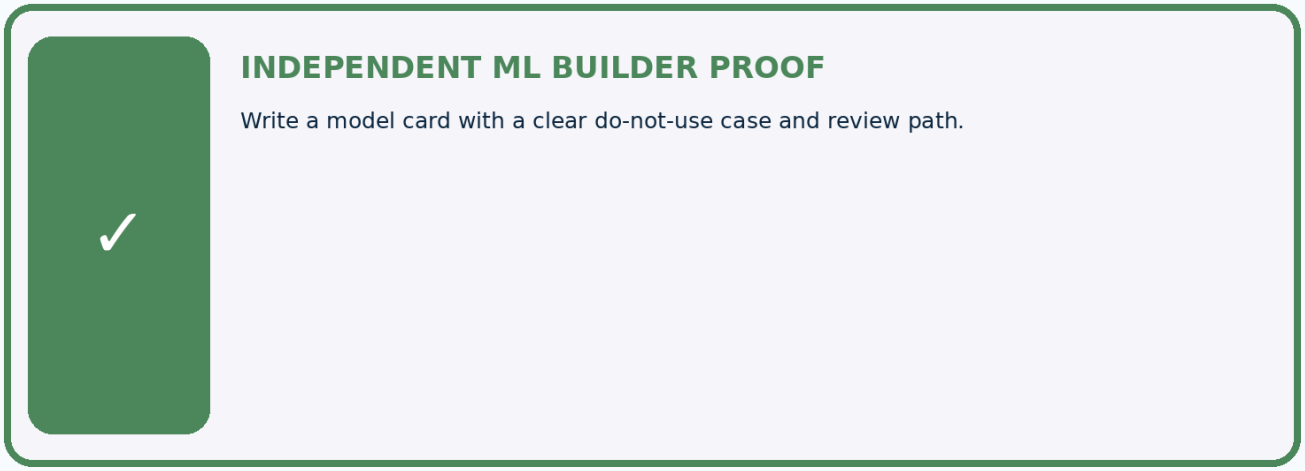
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Write a **model card** with a clear **do-not-use** case and review path.

NEXT BRIDGE Carry the experiment **evidence** into: Capstone Dataset and Baseline.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A **model card** says healthy/spotted,**80%** accuracy only.

Add responsible-use **evidence**.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Intended users/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and **human review**. Classroom classification is not professional diagnosis.

Evidence or stage	Value or result
Purpose	Users, conditions and excluded uses
Evidence	Data, labels, metrics and subgroup results
Version	System and evaluation identity
Limits	Failure conditions and human review

▼ Earlier wording, alternatives and picture transcripts

Week 33 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A model card supports responsible review

A **model card** says healthy/spotted,**80%** accuracy only.

Add responsible-use **evidence**.

Intended users/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and **human review**. Classroom classification is not professional diagnosis.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Write a model card whose claims match evidence	Explain the deciding step, show a trace or test, and state one limit.
Separate intended use from unsupported use	Explain the deciding step, show a trace or test, and state one limit.
Review a card for missing evaluation details	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Write a **model card** whose performance statement can be traced to a concrete evaluation.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Write a **model card** whose performance statement can be traced to a concrete evaluation.

- Which labelled step matches this detail? Do not claim all cameras, outdoor conditions or disease diagnosis were validated.
- 1. Name **evidence**
- 2. State limits
- 3. Compare baseline
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: State limits Do not claim all cameras, outdoor conditions or disease diagnosis were validated. Read the picture in order.

Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The publicity sentence becomes "**80%** accurate for every plant". Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.
- 3. Reject that unbounded statement.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Reject that unbounded statement. The measured rate belongs to twenty related indoor photos, not every plant or setting.

Repair and keep the record

Review Proxy Features and Fairness (ai-4/32). Goal: Identify possible proxy features. Short practice: What **evidence** contradicts the original “all setting information removed” claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Improve Generalization (ai-4/30). Goal: Plan one **evidence**-led change and test it on source-separated validation cases. Short practice: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Over**fitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

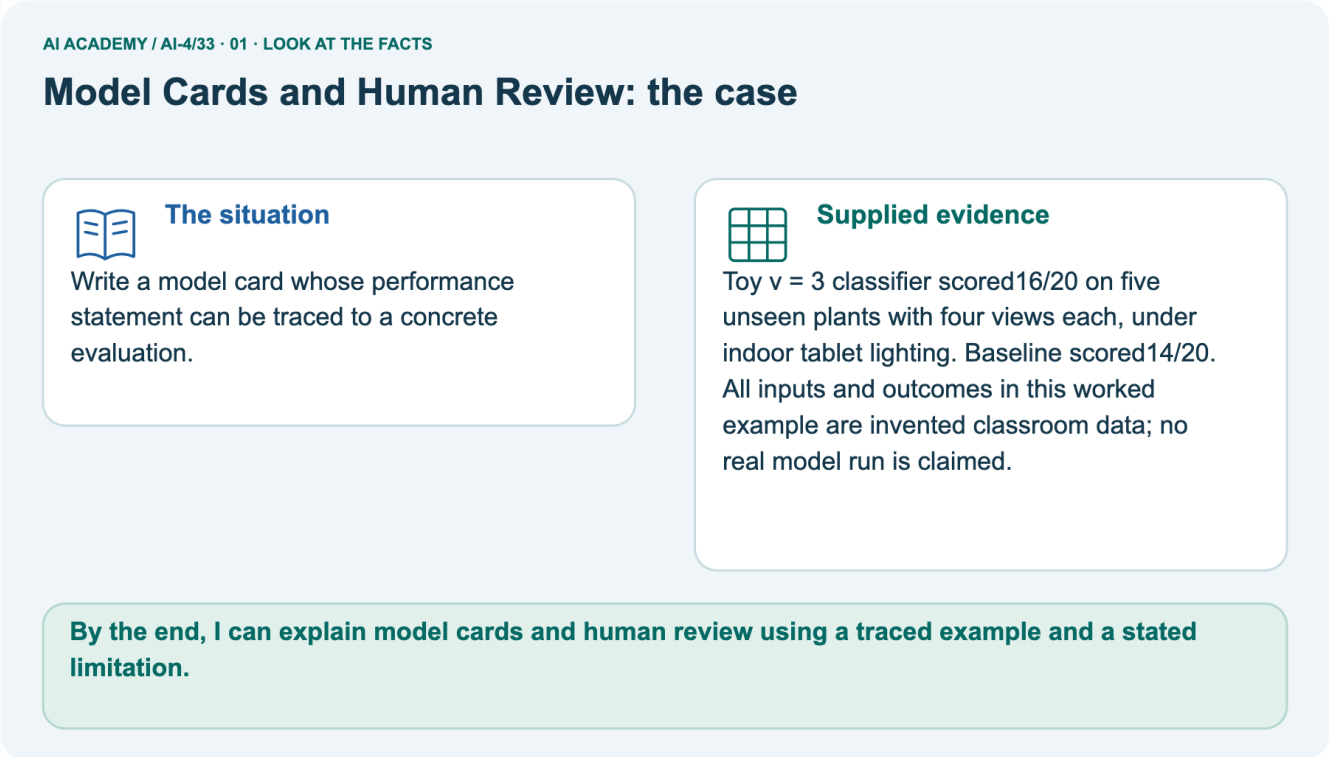
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Model Cards and Human Review: the case



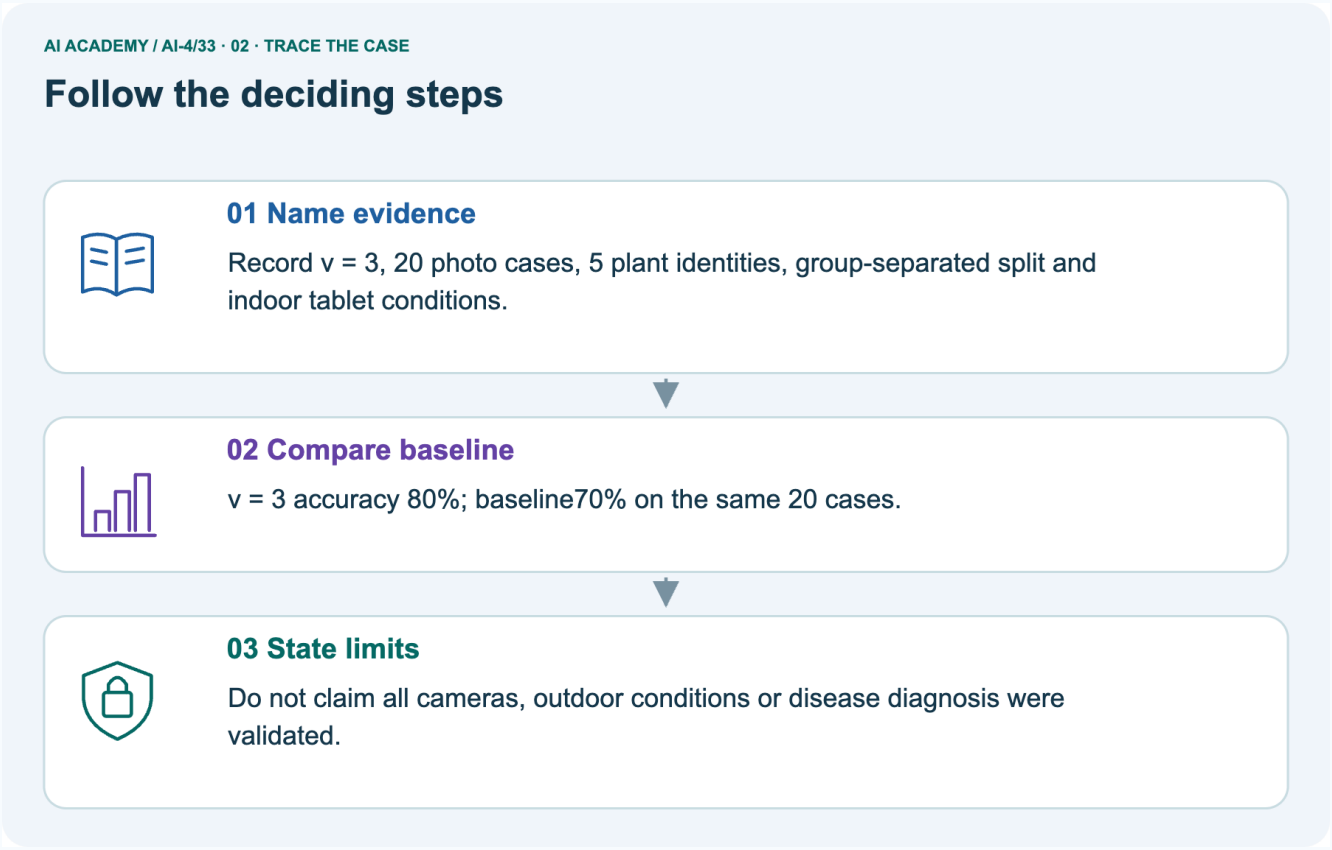
Follow the labelled picture with the matching explanation. Open the vector diagram

Write a **model card** whose performance statement can be traced to a concrete evaluation. Supplied facts: Toy v = 3 classifier scored16/20 on five unseen plants with four views each, under indoor tablet lighting.

Baseline scored14/20. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

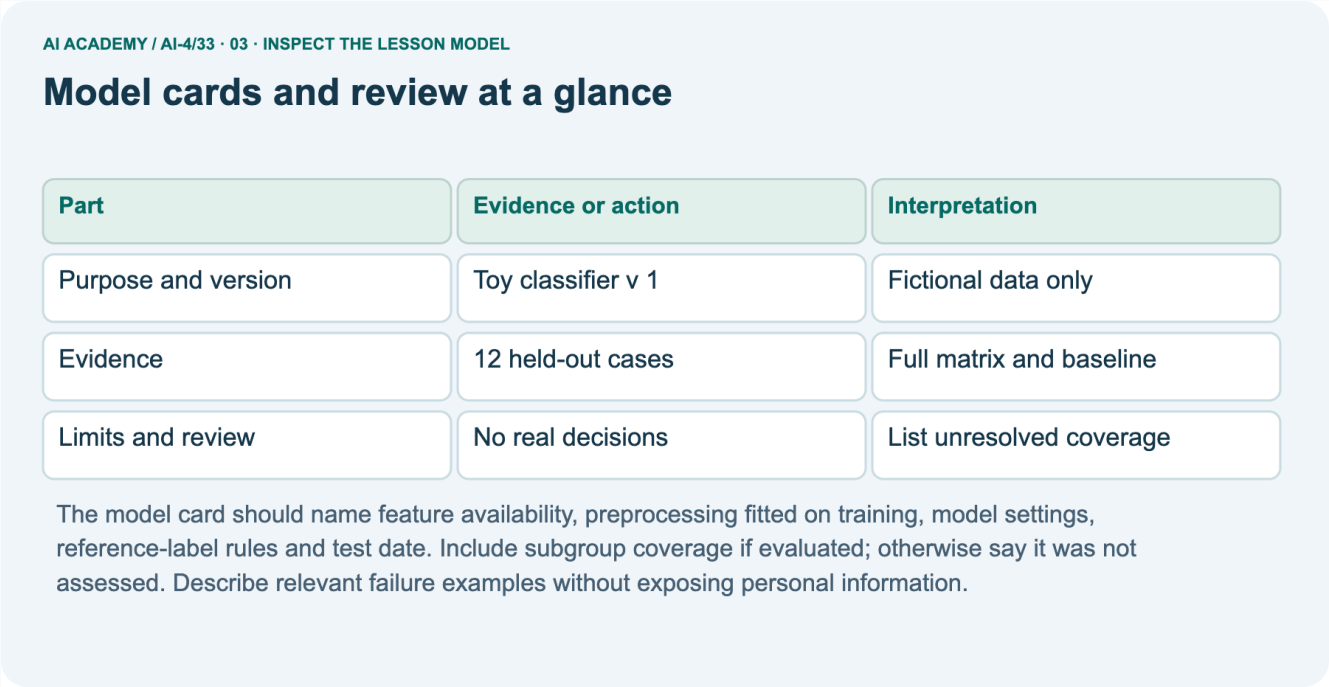


Follow the labelled picture with the matching explanation. Open the vector diagram

Name **evidence**: Record v = 3, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions. → Compare baseline: v = 3 accuracy **80%**; baseline**70%** on the same 20 cases. → State limits: Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

Point to each numbered stage and explain which recorded **input** it uses.

03. Model cards and review at a glance

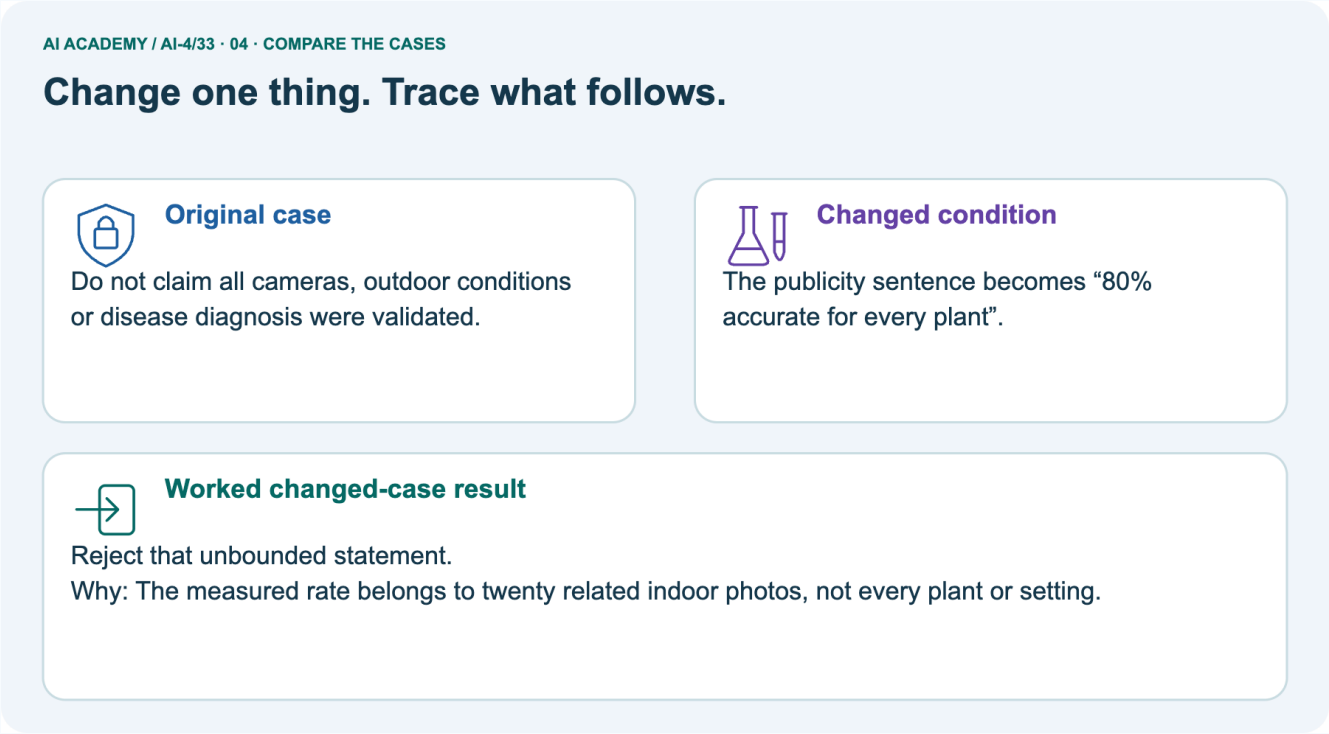


Follow the labelled picture with the matching explanation. Open the vector diagram

Model cards and review at a glance. Purpose and version | Toy classifier v 1 | Fictional data only. **Evidence** | 12 held-out cases | Full matrix and baseline. Limits and review | No real decisions | List unresolved coverage

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



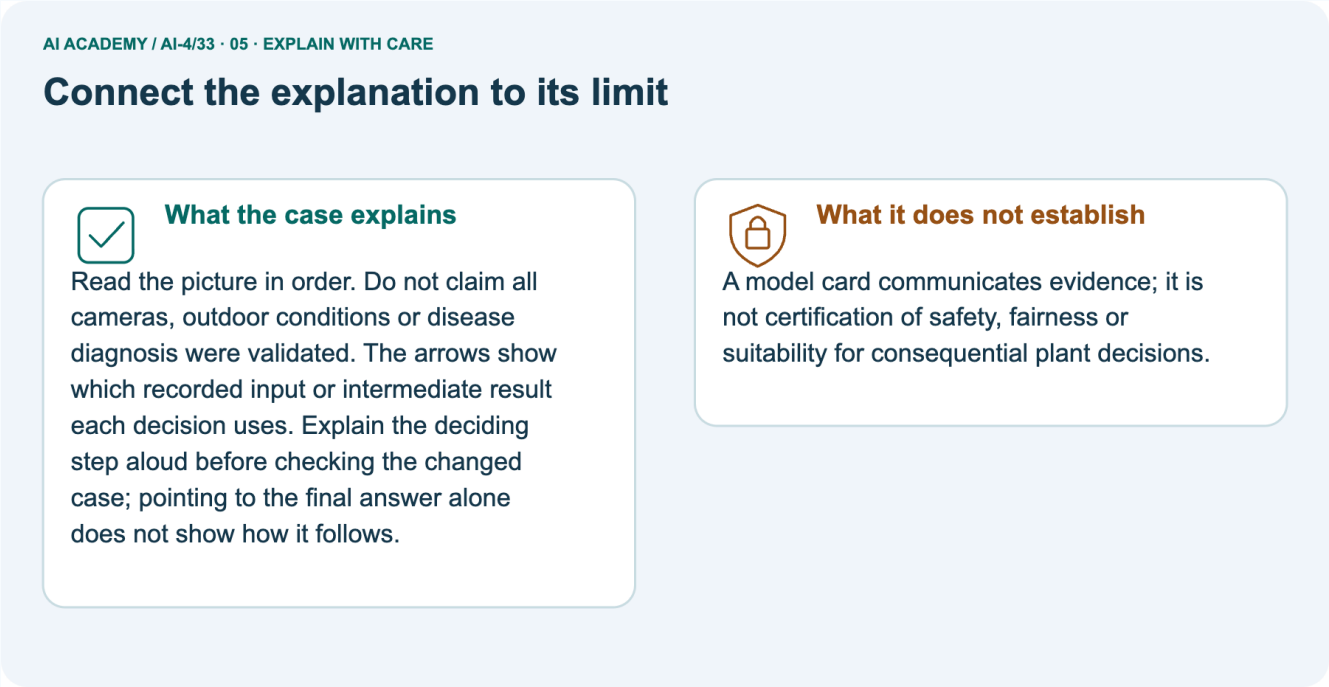
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Do not claim all cameras, outdoor conditions or disease diagnosis were validated. Changed condition: The publicity sentence becomes “**80%** accurate for every plant”.

Expected result: Reject that unbounded statement. Reason: The measured rate belongs to twenty related indoor photos, not every plant or setting.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

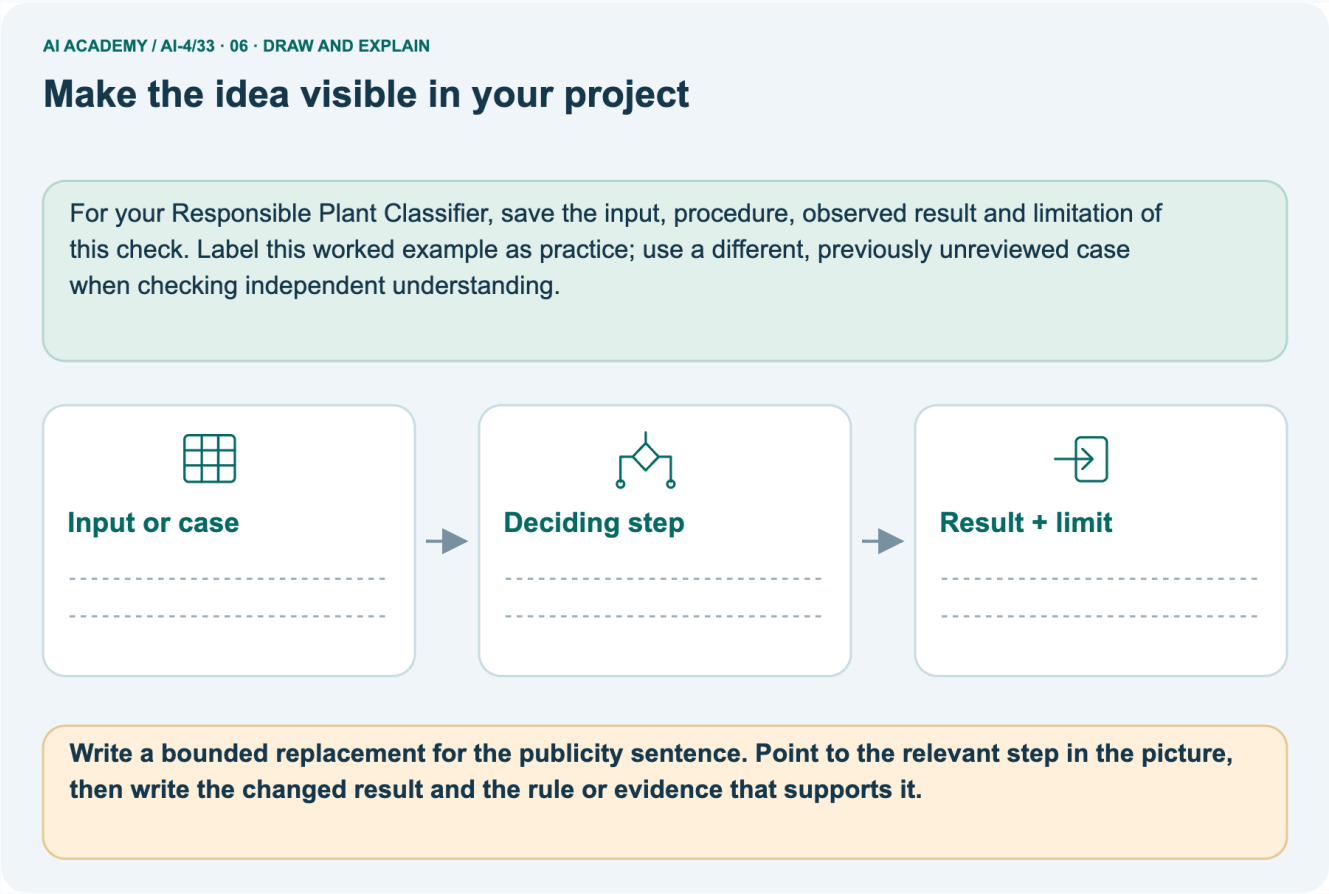
Read the picture in order. Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **model card** communicates **evidence**; it is not certification of safety, fairness or suitability for consequential plant decisions.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Model Cards and Human Review: the case

02. Follow the deciding steps

03. Model cards and review at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.